High Level Design (HLD)

Project Title

Design & Build a RAG-Based Customer Support Assistant

System Overview

AI-powered customer support assistant using Retrieval-Augmented Generation to answer queries from a PDF knowledge base.

Architecture Components

- User Interface (Streamlit)
- PDF Loader
- Chunking Module
- Embedding Model
- Vector Database
- Retriever
- LLM Response Engine
- LangGraph Workflow
- HITL Module

Data Flow

PDF -> Chunking -> Embeddings -> Vector Store -> User Query -> Retrieval -> Answer Generation -> Output

Technology Choices

Python, Streamlit, LangChain, LangGraph, FAISS/ChromaDB, Gemini API

Scalability

Supports larger documents, more users, and future cloud deployment.